

SUNGMIN LEE

Semiconductor AI Engineer | Full-Stack Developer | Semiconductor Data Expert

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Summary

Semiconductor AI Engineer with experience designing, developing, and operating AI/Data systems for real semiconductor engineering workflows.

- **Production AI for Semiconductor Engineering:** Developed and operated practical AI systems for semiconductor engineering workflows, including image anomaly detection and RAG-based search platforms.
- **Large-Scale Semiconductor Data Processing & Platform Development:** Developed and operated end-to-end platforms that collect, preprocess, analyze, and serve various semiconductor data such as lot history, images, process recipes, and reports.
- **Semiconductor Data Domain Expertise:** Performed DRAM product defect analysis and major lot quality management to support yield improvement.
- **Leadership & Capability:** Experienced in development organization operation, project planning, development policy setup, development leadership, architecture design, cross-project integration, operation framework setup, and rapid application of new technologies to engineering workflows.

Professional Experience

Samsung Electronics Co., Ltd.

Hwaseong-si, Republic of Korea

Staff Engineer, Evaluation & Analysis Process Development

2022.03 - present

- Developed and operated practical end-to-end AI systems for semiconductor engineering workflows.
- Built clusters, databases, data pipelines, and security environments for semiconductor data such as lot history, metrology images, process recipes, and reports.
- Developed full-stack applications and operated infrastructure environments for AI/Data systems, including servers, GPUs, CI/CD, MLflow, and Airflow.
- Led multiple AI/SW projects, operated a development squad, planned department-level projects, and established development policies.

Samsung Electronics Co., Ltd.

Hwaseong-si, Republic of Korea

Staff Engineer, Evaluation & Analysis Process Development

2018.11 - 2022.02

- Performed defect analysis for DRAM product development projects, including D1z, D1a, D1b, and D1c.
- Automated repetitive RCA and data analysis workflows.
- Managed the health status of major lots by tracking and predicting process history, metrology, inspection, events, schedules, and lot status.
- Conducted metrology anomaly detection activities.

Hyundai Mobis Co., Ltd.

Yongin-si, Republic of Korea

Research Engineer, Steering System Engineer Team

2017.07 - 2018.10

- Developed and evaluated automotive steering systems.
- Performed gear design optimization.

Main Projects

CADI: Image Anomaly Detection Platform

2026.01 - 2026.06

Developed an AI anomaly detection platform to identify image-based defects in metrology images that are difficult to detect using conventional statistical or numerical monitoring. The core idea was to detect anomalies by comparing relative differences within image groups captured under the same pattern and alignment conditions. Through this approach, the platform delivered a generic anomaly detection algorithm and system applicable to new products and new defect types without model retraining.

- **Key Contributions:** Development lead and main developer; designed and tested a generic anomaly detection algorithm; implemented multi-GPU inference and end-to-end system development.
- **Impact:** Improved recall from **67% to 90%** compared with PatchCore; improved throughput by **40×** compared with single-thread processing; estimated **30 person-months** of workload reduction; enabled image anomaly detection that was previously difficult with conventional monitoring; developed a generic anomaly detection algorithm.

DAS: RCA Data Analysis Platform

2022.03 - 2023.10

Developed an RCA data analysis platform to support defect root cause analysis in a semiconductor R&D environment. The platform addressed challenges such as limited sample size and branched process history by applying data preprocessing logic and statistical quantification models. It integrates various manufacturing data, including process, equipment, chamber, reticle, metrology, inspection, overlay, VM, FDC, and delay-time data, to quantitatively identify candidate root causes and support engineers in interpreting analysis results.

- **Key Contributions:** Sole developer; performed end-to-end ownership from planning and development to operation.
- **Impact:** Estimated **40 person-months** of workload reduction; established as a representative RCA analysis platform in the semiconductor R&D center; provided a self-service environment where non-expert engineers can perform structured RCA analysis.

ARISE & Code Platform: RAG Search Platforms

2025.03 - 2025.10

Developed RAG-based search platforms to improve accessibility and reuse of semiconductor engineering knowledge assets. ARISE is a platform for searching PPT reports such as defect reports, analysis reports, and material analysis documents. Code Platform is a platform for searching SQL queries and Python scripts used in semiconductor data analysis. Previously, these assets were managed separately by individuals or departments, making reuse difficult, and keyword-based search was not sufficient to find relevant information. To solve this problem, designed and implemented an adaptive RAG architecture.

- **Key Contributions:** Project lead and main developer; designed vector DB schemas for structuring data and metadata; designed and implemented adaptive RAG architecture for search; performed QA evaluation; implemented SSE-based response streaming.
- **Impact:** Improved accessibility and reusability of semiconductor knowledge and engineering assets; ARISE improved search accuracy from **60% to 90%** compared with naive RAG and delivered an estimated **30 person-months** of workload reduction; Code Platform improved search accuracy from **20% to 70%** compared with naive RAG and delivered an estimated **30 person-months** of workload reduction.

AQUA: Process Recipe Platform

2023.03 - present

Developed a database and web platform to collect and standardize CLN recipe data that had been individually managed by engineers due to security and operational constraints. The platform improves accessibility to secured recipe data and enables users to retrieve actual process recipe information. It also integrates recipe information, process history, and evaluation data by leveraging metrology data and ER rate information.

- **Key Contributions:** Main developer; addressed recipe security and accessibility issues; designed and built data pipelines for recipe, process history, and evaluation data.
- **Impact:** Estimated **30 person-months** of workload reduction; improved accessibility to secured recipe data; integrated recipe, history, and evaluation information into a unified platform.

Education

Seoul National University

Seoul, Republic of Korea

M.Sc., Department of Naval Architecture and Ocean Engineering

2015.09 - 2017.06

- **Major:** Optimization Algorithm
- **Thesis:** Optimization of Ship Route Planning

Seoul National University

Seoul, Republic of Korea

B.S., M.Sc., Department of Naval Architecture and Ocean Engineering

2009.03 - 2015.08

Additional

- **Awards, Samsung Electronics Internal:** 2025 AI BP Festival / 2024 SRD e-Technology Journal / 2023 The Innovator
- **Teaching, Samsung Electronics Internal:** Spotfire Workshop (2024–Present) / Practical ETL Course (2024–2025) / Fundamentals of Data Analysis (2024–2025)
- **Publication:** Spotfire 간단히 그래프 그리기, ISBN: 9788924098136, 2022.07

Technical Skills

- **AI/ML:** PyTorch, Scikit-learn, LangGraph
- **Backend:** Python, FastAPI
- **Frontend:** Vue.js, React
- **Infrastructure:** Linux, GPU, Docker, Kubernetes, Git, NAS
- **Database:** RDBMS, NoSQL, Vector DB, Graph DB
- **BI:** Spotfire
- **Semiconductor Domain:** DRAM Development, Root Cause Analysis, Defect Analysis, MI